ANDREW TSENG

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EDUCATION

University of Texas at Austin **Graduation Date: December 2018**
Bachelor of Science in Computer Science

WORK EXPERIENCE

Microsoft - Software Engineer II - Windows Cloud EXP December 2021 - August 2026

- Full stack developer for the first party "Windows App" application (aka.ms/WindowsApp)
- Internal build and release expert for the Microsoft Windows App and Remote Desktop application
- Integrated agentic development into triage cycle. Connected ICM system to triage agent to fast track investigation on all issues and bugs.
- Worked across teams and designers to integrate new UX flows for RDP connections within Windows App.
- Assisted sister teams with recurring issues and bugs when needed. Led learnings on relevant tools and processes to fast track developers into contributing to features.

Kershner Trading Group - Software Engineer February 2019 - December 2021

- Improved the Locate Borrow application by removing C# dependencies within the C++/CLR application. Added features in application UI to provide visual TCP socket statuses for DevOps staff.
- Developed server to handle orders for new clients and translates FIX 4.2 messages to an internal message.
- Developed test client to test fix server that takes in FIX 4.2 messages to pipeline in to order handling system.
- Ported code from windows to Linux to fit new feed servers/hardware lead to improve latency in handling orders.
- Built internal tools to monitor system of order management system and found 21 inter-application bugs.
- Designed and developed backend feature of recycling shares within trader accounts for to use throughout trading day.

Compal Electronics - Software Engineer Intern Summer, 2016

- Rewrote developer admin database pages into templates and added management features.
- Developed sanity tests to help with continuous integration to help find bugs in the in-house API.
- Debugged and refactored MongoDB schema for new category of data in database.
- Researched IoT concepts/protocols for R&D for potential uses on new products.

PROJECTS

Type Inference System in C

- Implemented type inference features in C with the **auto** keyword.
- Created symbol table and file rewrite in replacing auto keyword to correct type.
- Rewrites the auto keyword to the type that it infers based on the C-program.
- Wrote test scripts in Bash to make find corner cases along with string parser that finds lexes within the input program.

Linux Kernel

- Developed Linux kernel in C++ featuring threads, concurrency tools, virtual memory, file systems and system calls.
- Designed threading system in intention to prevent deadlocking and manage re-entrant functions.
- To avoid baggage that C++ brings, avoided using C++ STL throughout development.

Lilypond - Facebook Messenger Bot

- Facebook Messenger bot that converts textual music notes into a music sheet.
- Utilized ImageMagick, C++, NodeJS to convert and parse text and create music sheet PNG file.

LANGUAGES AND TOOLS

Experience : Python | Java | C/ C++ | Visual Studio | Git | Grafana | Azure DevOps | WinDBG
Familiar : MySQL | R | LaTeX| Docker | Google Cloud Platform | Javascript/React | Jenkins | Anaconda | Microsoft SQL MS |